

Exploratory Data Analysis (EDA) of E-Commerce Sales Data

Project Overview

Background

As part of my internship program, I was tasked with performing an **Exploratory Data Analysis (EDA)** on an e-commerce sales dataset. The objective was to examine the dataset, identify patterns, understand customer purchasing behavior, detect anomalies, and uncover meaningful business insights through statistical and visual analysis.

EDA is a critical stage in the data analysis process because it helps analysts understand the structure of the data before applying advanced analytical or predictive techniques.

Project Objectives

The primary objectives of this project were to:

- Calculate and interpret basic descriptive statistics.
 - Analyze the distribution of key business variables.
 - Identify trends in sales and order activity over time.
 - Detect potential outliers within the dataset.
 - Summarize key findings and business insights.
 - Present results through clear visualizations and dashboards.
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Tools Used

- Microsoft Excel
 - Pivot Tables
 - Pivot Charts
 - Statistical Functions
 - Dashboard Design
 - Data Visualization

Skills Demonstrated

- Exploratory Data Analysis (EDA)
- Descriptive Statistics
- Trend Analysis
- Distribution Analysis
- Outlier Detection
- Business Insight Generation
- Data Visualization
- Dashboard Development
- Data Storytelling

Dataset Description

The dataset contains transactional information from an e-commerce platform, including:

Feature	Description
Order ID	Unique identifier for each order
Customer ID	Unique identifier for customers
Products	Product purchased
Quantity	Number of items purchased
Unit Price	Price per item
Total Price	Total order value
Payment Method	Method used for payment
Coupon Code	Promotional coupon applied
Referral Source	Customer acquisition source
Order Status	Current order status
Order Date	Date of transaction

Metric	Value
Total Quantity Sold	3,535
Total Revenue	1,264,761.96
Total Orders	1,200
Total Customers	1,189

Quantity Statistics

Statistic	Value
Average Quantity	3
Minimum Quantity	1
Maximum Quantity	5

Unit Price Statistics

Statistic	Value
Average Unit Price	356.41
Lowest Price	11.39
Highest Price	699.93

Cart Statistics

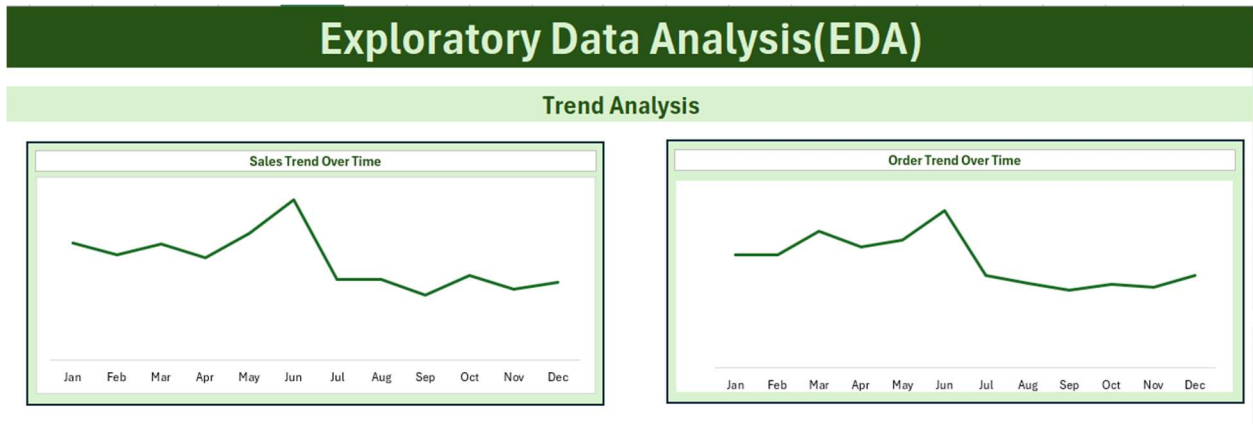
Statistic	Value
Average Items per Cart	5
Minimum Items per Cart	1
Maximum Items per Cart	10

Order Value Statistics

Statistic	Value
Average Order Value	1,053.97
Minimum Order Value	11.39

Statistic	Value
Maximum Order Value	3,456.40

Trend Analysis



Sales Trend Over Time

The sales trend displayed moderate fluctuations throughout the years.

Key Findings

- Sales increased steadily from January to June YoY.
- Peak sales occurred always in June.
- A noticeable decline was observed after June.
- Sales remained relatively stable during the second half of the years.

Business Interpretation

The sales peak may indicate seasonal demand, successful promotional campaigns, or increased customer activity during mid-year periods.

Order Trend Over Time

The order volume trend closely mirrored the sales trend.

Key Findings

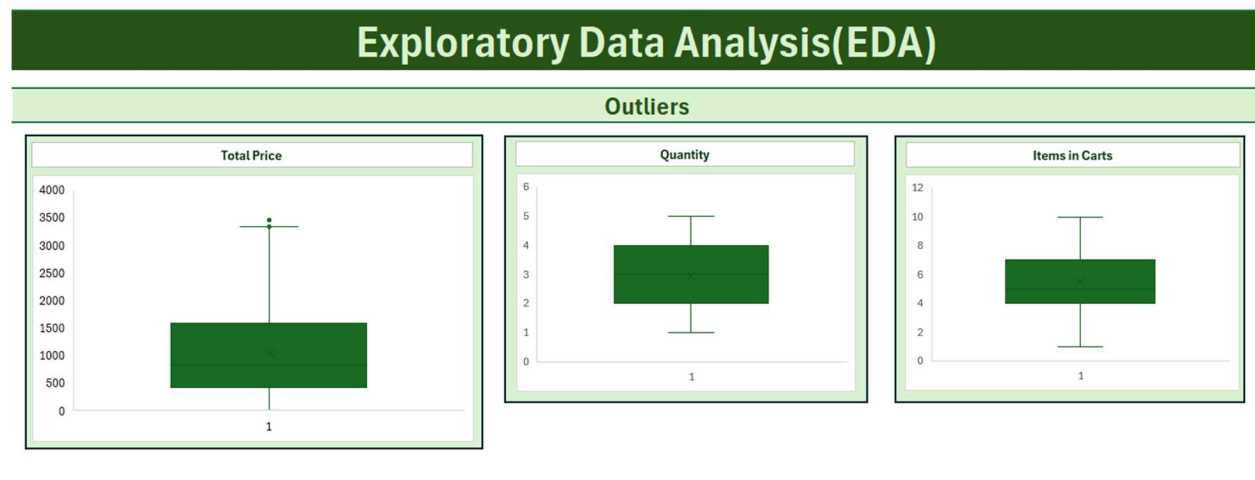
- Order activity increased during the first half of the year.

- June recorded the highest order volume.
- Orders declined significantly in July.
- Activity stabilized between August and December.

Business Interpretation

The alignment between sales and order volume suggests that revenue growth is primarily driven by increases in transaction volume rather than price changes.

Outlier Analysis



Boxplots were used to identify unusual observations within key numerical variables.

Total Price

The Total Price distribution exhibited several high-value outliers.

Interpretation

- A small number of orders were significantly larger than the average order value.
- These transactions may represent bulk purchases or premium product purchases.
- Such orders positively impact overall revenue but can distort average-based metrics.

Quantity

The Quantity variable showed no significant outliers.

Interpretation

- Customer purchase quantities remain within a consistent and expected range.
- Most customers purchase between 1 and 5 items per transaction.

Items in Cart

The Items in Cart variable displayed a relatively normal distribution without extreme values.

Interpretation

- Shopping cart behavior is consistent across customers.
- No unusual purchasing patterns were observed.

Distribution Analysis



Product Distribution

The product categories generated relatively balanced revenue contributions.

Key observations:

- Chairs generated the highest revenue contribution (~195,620).
- Printers closely followed (~195,613).
- Laptops contributed approximately 192,127.

- Phones recorded the lowest revenue contribution (~151,722).

This suggests that revenue generation is diversified across multiple product categories rather than being dependent on a single product line.

Payment Method Distribution

The analysis showed a fairly even distribution across payment methods:

Payment Method	Orders
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Insight

Online payments were the most preferred method, indicating strong customer adoption of digital payment channels.

Coupon Usage Analysis

Coupon	Usage Count
FREESHIP	313
None	309
WINTER15	292
SAVE10	286

Insight

- FREESHIP was the most frequently used coupon.
- Coupon usage was relatively balanced.

- A significant number of customers completed purchases without promotional discounts.

Referral Source Distribution

Source	Customers
Instagram	259
Email	250
Google	241
Facebook	228
Referral	222

Insight

Instagram emerged as the strongest customer acquisition channel, indicating the effectiveness of social media marketing.

Order Status Distribution

Status	Count
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Insight

The distribution of order statuses appears balanced. However, the relatively high number of cancelled and returned orders may require further investigation to identify potential operational or customer satisfaction issues.

Key Business Insights

Based on the analysis, the following insights were identified:

Revenue Performance

- The business generated over **1.26 million** in total revenue.
- Average order value exceeded **1,000**, indicating strong transaction values.

Customer Behavior

- Customers typically purchase approximately **3 items per order**.
- The average shopping cart contains **5 items**.

Product Performance

- Printers, chairs, and laptops were the strongest revenue-generating product categories.
- Phone sales generated comparatively lower revenue.

Marketing Performance

- Instagram was the most effective referral channel.
- Email and Google also contributed significantly to customer acquisition.

Payment Preferences

- Online payment methods were slightly more popular than traditional payment options.
- Digital payment adoption appears strong among customers.

Operational Considerations

- High counts of cancelled and returned orders may indicate opportunities for process improvement.
 - Further analysis should be conducted to determine root causes.
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Conclusion

This Exploratory Data Analysis successfully provided a comprehensive understanding of the e-commerce dataset by examining descriptive statistics, distributions, trends, and outliers.

The analysis revealed that sales and order activity peaked during the middle of the year, digital payment methods were widely adopted, Instagram served as the leading acquisition channel, and product revenue was distributed across multiple categories. While customer purchasing behavior appeared stable, the number of cancelled and returned orders suggests an area for further investigation.

Overall, the project demonstrates how EDA can be used to transform raw transactional data into meaningful business insights that support data-driven decision-making.